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Integration and Validation of Whole-Body Model Predictive Control for a Mobile Manipulator in Line-less Assembly

Integration und Validierung einer ganzkörperbasierten modellprädiktiven Regelung für einen mobilen Manipulator in der linienlosen Montage

Project Thesis / **Projektarbeit**

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Introduction to Line-less Mobile Assembly Systems

1.1 Line-less Mobile Assembly Systems and Their Implications for Motion Planning

Manufacturing is increasingly shaped by short product life cycles, high variant diversity, and volatile demand. Rigid, station-based assembly lines are optimised for a fixed product mix and a stable takt time. When the product or the required volume changes, the layout, material flow, and station balancing must be reconfigured, which is slow and capital-intensive [HÜTT16]. The flexibility that volatile demand requires and the rigidity of a fixed line pull in opposite directions, which has motivated alternative production paradigms.

The *Line-less Mobile Assembly System* (LMAS) is one such paradigm studied at the Laboratory for Machine Tools and Production Engineering (WZL) of RWTH Aachen University [KERN19; BUCK22]. Instead of fixing products, tools, and workers to a sequence of static stations, an LMAS treats all production resources—mobile robots, mobile manipulators, workpiece carriers, and human workers—as *mobile, freely relocatable entities* in a shared shop floor. The assembly "line" is dissolved into a dynamically reconfigurable constellation of resources that are matched to the current order at run time. This yields a spatially and temporally flexible production system that can absorb product changes, demand fluctuations, and disturbances without a physical relayout.

A central enabler of the LMAS concept is the *mobile manipulator*: a robotic arm mounted on a mobile base. The mobile base provides an effectively unbounded workspace and the ability to reposition itself between and within tasks, while the arm provides the dexterity required for the assembly operation itself. Combining the two allows a single resource to service many workstations and to reach parts of a large workpiece that no fixed-base arm could cover.

This freedom changes the motion-planning problem. In a classical line, an industrial arm operates in a carefully engineered, static cell: the part arrives at a known pose, the surroundings are fixed, and the arm executes largely pre-programmed trajectories. In an LMAS, several assumptions no longer hold:

- **Large, base-dependent workspace.** The target end-effector pose may lie outside the reach of the arm from the current base pose, so the base and the arm must *cooperate* to accomplish a single task.
- **Changing, shared environment.** Resources move relative to one another; the environment is neither static nor fully known in advance, so trajectories cannot be fully pre-computed offline.
- **Kinematic redundancy.** A mobile base combined with a six-degree-of-freedom (6 DOF) arm is highly redundant with respect to an end-effector task. This redundancy is an opportunity—it can be exploited for reach, manipulability, or obstacle avoidance—but only if the planner reasons about the base and the arm *jointly*.

Together, these conditions call for motion planning that is *reactive*, that spans the *combined* configuration space of base and arm, and that resolves the resulting redundancy deliberately rather than as a by-product. Whole-body motion planning, introduced next, is aimed at exactly this problem.

1.2 The Need for Whole-Body Motion Planning: From Sequential to Unified Approaches

The most common way to command a mobile manipulator is to treat the base and the arm as two independent subsystems and to plan for them *sequentially*: a navigation stack first drives the base to a suitable stand-off pose, and only then does an arm-motion planner (for example, a sampling-based planner behind MoveIt [PICK24]) compute a trajectory for the arm from that fixed base pose. This decoupled scheme is simple, modular, and reuses mature, well-tested components, which explains its prevalence in industry.

The decoupling, however, has intrinsic limitations that become severe in the LMAS setting:

- **Loss of optimality and reach.** Choosing the base pose *before* planning the arm discards the coupling between the two subsystems. A base pose that looks reasonable geometrically may leave the arm near a singularity or a joint limit, or may make the end-effector target unreachable, even though a jointly chosen base–arm configuration exists.

- **Stop-and-go behaviour.** Sequential execution—drive, stop, then manipulate—prevents the base and the arm from moving *simultaneously*. Tasks in which the end-effector must follow a continuous path longer than the arm's reach (a common LMAS case) cannot be executed smoothly this way.
- **Poor disturbance rejection.** A plan computed once, for a fixed base pose, cannot react to base drift, moving obstacles, or an updated target without re-triggering the whole pipeline.

Whole-body motion planning removes this artificial boundary by treating the mobile base and the arm as a *single* redundant kinematic (or dynamic) system and planning their motion in one unified optimisation. The end-effector task is expressed directly in task space, and the planner is free to distribute the required motion between the base and the arm according to a chosen cost criterion. This directly addresses the limitations above: the full combined workspace becomes available, base and arm move simultaneously along one coordinated trajectory, and, once embedded in a feedback loop, the plan reacts continuously to a changing target and environment.

Formulating whole-body planning as a receding-horizon optimal-control problem, i.e. as *Model Predictive Control* (MPC) [RAWL17], is well suited to mobile manipulation. At each control step the planner solves a finite-horizon optimisation over the whole-body state and input, applies the first part of the resulting policy, and re-plans as new state estimates arrive. Constraints such as joint and velocity limits, self-collision, and the non-holonomic behaviour of a differential-drive base are handled inside the same optimisation, and the receding-horizon re-planning provides the reactivity that the LMAS setting requires. The open-source OCS2 (Optimal Control for Switched Systems) framework [FARS17] provides a mature implementation of this approach and has been demonstrated for whole-body loco-manipulation on legged and wheeled platforms [SLEI21]. It is the framework integrated and evaluated in this work.

1.3 Objectives and Structure of This Work

Objective.

The goal of this work is to *integrate and validate* whole-body MPC, based on the OCS2 framework, on the HERo mobile manipulator—a differential-drive mobile base carrying a Universal Robots UR10e arm [UNIV24]—and to characterise the resulting end-effector-pose tracking behaviour under realistic conditions. In contrast to studies that remain in simulation, the emphasis here is on a *deployable* system: the MPC runs in a distributed, real-time software architecture on the physical robot, and its performance is assessed on hardware.

Scope and contributions.

The concrete contributions of this thesis are:

1. the selection of a suitable whole-body motion-planning approach for the HERo platform, following a comparison of candidate frameworks (Chapter 2);
2. the integration of OCS2 whole-body MPC into the HERo software stack, including a distributed host/on-board architecture for real-time operation, the whole-body model and its parameterisation, and the hardware and ROS-control interfaces (Chapter 3);
3. a simulation-based validation of the closed loop that isolates the geometry, timing, and data pipeline before hardware deployment (Chapter 4); and
4. the deployment of the system on the physical HERo platform, together with a discussion of the deployment challenges encountered and preliminary end-effector-tracking results (Chapter 5).

Structure.

The remainder of this thesis is organised as follows. Chapter 2 reviews the state of the art in whole-body motion planning for mobile manipulators, classifies the main approaches, presents OCS2 and alternative frameworks, and justifies the selection made here. Chapter 3 describes the HERo platform and the integration of OCS2 into it: the distributed architecture, the whole-body model and its parameterisation, and the interfaces to the low-level control layer. Chapter 4 presents the simulation-based validation using a kinematic closed-loop (dummy MRT) setup. Chapter 5 reports on the hardware deployment, the challenges and their mitigations, and preliminary end-effector-tracking results. Chapter 6 summarises the work, reflects critically on its limitations, and gives recommendations for resolving the remaining issues.

2

State of the Art in Whole-Body Motion Planning for Mobile Manipulators

The previous chapter argued that the Line-less Mobile Assembly System (LMAS) setting calls for whole-body motion planning rather than a decoupled base–arm pipeline. This chapter surveys how the literature approaches the problem and narrows the field down to the framework used in the remainder of this thesis. Section 2.1 introduces a classification of motion-planning approaches for mobile manipulators. Section 2.2 then focuses on *holistic* (whole-body) planners, presenting the OCS2 framework (Section 2.2.1) and the main alternatives (Section 2.2.2). Section 2.3 compares the candidates against the requirements of the HERo platform and justifies the selection of OCS2. The synthesis of the applied and open-source landscape draws, in part, on a systematic literature review of whole-body motion planning and control integration in robotic navigation stacks carried out within the same research project [SANT26].

2.1 Classification of Motion Planning Approaches

Motion planning for mobile manipulators can be organised along a few largely orthogonal axes. Systematic reviews of the field [SAND22; THAK23] distinguish approaches primarily by *how the base and the arm are coupled* and by *the algorithmic principle* used to generate motion; a third, practical axis concerns *where the computation happens in time.*

Decoupled versus whole-body.

The first and, for this work, most important distinction is between *decoupled* and *whole-body* schemes. A decoupled scheme plans the base and the arm separately and in sequence: a navigation stack drives the base to a stand-off pose, after which an arm planner solves the manipulation task from that fixed base configuration. A whole-body scheme instead treats the base and the arm as a *single* redundant

kinematic (or dynamic) system and solves for their motion jointly, so that the task-space objective is distributed between base and arm by the planner itself. As motivated in Section 1.2, decoupling sacrifices reach, optimality, and reactivity, which is what whole-body planning recovers, at the cost of a higher-dimensional and more tightly coupled problem.

Algorithmic principle.

Independently of the coupling, the motion itself can be produced by different families of algorithms:

- **Sampling-based planning** explores the configuration space by randomly sampling and connecting collision-free states (for example Rapidly-exploring Random Trees and Probabilistic Roadmaps, as exposed through MoveIt [PICK24]). These methods excel at finding feasible paths in cluttered, high-dimensional spaces but do not natively optimise a running cost and are typically used offline.
- **Optimization-based planning** formulates motion as a constrained optimisation over a trajectory. When the optimisation is solved once, offline, over a full trajectory, one speaks of *trajectory optimization*; when it is re-solved online over a receding horizon, the same principle becomes *Model Predictive Control* (MPC) [RAWL17]. Both handle joint, velocity, and collision constraints explicitly and produce smooth, cost-optimal motion.
- **Reactive schemes** compute an instantaneous command from the current state, for example through potential fields or quadratic-program-based inverse kinematics (QP-IK). They are cheap and fast but myopic, lacking the look-ahead needed to respect long-horizon constraints.
- **Learning-based schemes** replace or augment the planner with a policy learned by (deep) reinforcement learning. They can amortise the online cost of whole-body planning but raise questions of data efficiency, safety, and sim-to-real transfer [SANT26].

Offline versus online.

Finally, a planner may operate *offline*—computing a complete trajectory before execution—or *online*, re-planning continuously as new state estimates and targets arrive. Only online, feedback-driven planning provides the disturbance rejection required in a shared, changing LMAS workspace. Whole-body MPC sits at the intersection of the *whole-body*, *optimization-based*, and *online* categories, which is the region of the design space this thesis targets.

2.2 Overview of Holistic Motion Planners

Holistic (whole-body) planners solve for base and arm motion in a single formulation. In practice, the mature, deployable representatives of this class are almost exclusively optimization-based, because expressing the task in a cost function and the physical restrictions as constraints maps naturally onto the coupled, redundant, and constrained nature of the mobile-manipulation problem. A systematic review of how such controllers are integrated with robotic navigation stacks found that, among the open-source projects that couple a navigation stack with a genuine whole-body controller, optimal-control formulations—and the OCS2 toolbox in particular—recur as a common backbone, alongside operational-space control, quadratic-program solvers, and, more recently, learned policies [SANT26]. The following subsections describe OCS2 and then position the principal alternatives against it.

2.2.1 The OCS2 Framework: Optimal-Control-Based Model Predictive Control

OCS2 (*Optimal Control for Switched Systems*) is an open-source C++ library for numerical optimal control developed at the Robotic Systems Lab, ETH Zürich [FARS17]. It provides efficient solvers for continuous-time optimal-control problems—most notably Sequential Linear-Quadratic (SLQ) control, a constrained variant of differential dynamic programming (DDP)—together with the supporting infrastructure required to run them as an MPC controller.

Three features make OCS2 well suited to whole-body mobile manipulation. First, it derives the cost and constraint derivatives that the solver needs by *automatic differentiation*: the user supplies the model (kinematics, cost terms, constraints) symbolically, and OCS2 generates and compiles efficient derivative code via CppAD/CppADCodeGen. Second, it exposes a *reference manager* that lets the desired task-space trajectory be updated online, which is exactly the interface a target-generating node needs in order to steer the controller. Third, and central to a real-time deployment, OCS2 separates the optimisation from the execution through an MPC–MRT split: the MPC process solves the optimal-control problem at its own rate and publishes a *policy*, while a lightweight Model Reference Tracking (MRT) client evaluates that policy at the control rate to produce the instantaneous command. This split makes it possible to run the optimiser and the hardware interface as separate processes, even on separate machines, a property this thesis exploits in the distributed architecture of Chapter 3.

OCS2 has been demonstrated for whole-body loco-manipulation on legged and wheeled platforms, including a unified MPC formulation for combined locomotion and manipulation [SLEI21] and perceptive whole-body interaction with articulated objects [MITT22]. Its lineage also includes foundational

work on kinematic planning for non-holonomic mobile manipulators and whole-body MPC for dynamically stable mobile manipulators, which the review in the same project identifies as the core mobile-manipulation references behind the toolbox [SANT26]. This track record on hardware, its explicit constraint handling, and its process-separable architecture make it the primary candidate for the HERo platform.

2.2.2 Alternative Frameworks and Methods

Several alternatives to OCS2 exist, spanning the algorithmic families of Section 2.1.

Sampling-based planning (MoveIt).

MoveIt [PICK24] is the de-facto standard motion-planning framework in the ROS ecosystem. Its strength is offline, collision-free planning for redundant arms in cluttered scenes; it can be extended to a mobile base by treating the base pose as additional planning variables. In its common usage, however, MoveIt underpins the *decoupled* navigate-then-manipulate pipeline and does not, by itself, provide the online, coordinated base–arm feedback that whole-body MPC offers. It is therefore the natural baseline against which a whole-body approach is compared rather than a competing whole-body planner.

Other optimization-based / MPC tools.

Beyond OCS2, whole-body behaviour can be realised with general-purpose optimal-control and MPC solvers or with constraint-based task-space controllers. The project-internal review catalogues open-source stacks that integrate such controllers with a navigation stack through different mechanisms—replacing the Nav2 local planner via the `nav2_core::Controller` plugin interface, wrapping the controller in `ros2_control`, or bridging `cmd_vel`/goal topics to a whole-body solver—and reports that OCS2-based and quadratic-program solvers dominate this category [SANT26]. These tools are technically capable but generally require more integration effort or offer less mature support for the specific combination of a differential-drive base and a ROS 2 hardware stack targeted here.

Reactive quadratic-program inverse kinematics.

QP-based inverse kinematics resolves the end-effector task and secondary objectives (joint-limit avoidance, manipulability) into joint velocities at each control step. It is computationally light and reactive but, being a one-step method, cannot anticipate constraints over a horizon, which limits its ability to plan coordinated base–arm motion for long, reach-exceeding paths.

Learning-based whole-body control.

A growing body of work hybridises classical navigation with reinforcement learning to amortise the online cost of whole-body planning, learning policies that map observations directly to base and arm commands [SANT26]. While promising, these approaches introduce training-data, reproducibility, and sim-to-real-transfer concerns that are not aligned with the goal of this thesis—deploying a transparent, analysable controller on real hardware in the near term.

2.3 Comparison of Motion Planning Algorithms and Selection of a Suitable Algorithm

The choice of a planner for HERo is driven by the requirements identified in Chapter 1: coordinated whole-body motion, explicit handling of joint/velocity, non-holonomic, and self-collision constraints, online reactivity, real-time feasibility on modest on-board compute, and good ROS 2 support so that the controller can be integrated with the existing hardware stack. Table 2.1 summarises how the candidate approaches score against these criteria.

Table 2.1: Qualitative comparison of motion-planning approaches against the requirements of the HERo platform (+ good, ◦ partial, – weak).

Criterion	Sampling (MoveIt)	QP-IK (reactive)	Learning (RL)	Whole-body MPC (OCS2)
Coordinated whole-body motion	◦	◦	+	+
Explicit constraint handling	◦	◦	–	+
Online reactivity	–	+	+	+
Real-time feasibility	◦	+	+	◦
Redundancy resolution	◦	+	+	+
ROS 2 maturity	+	◦	–	+
Transparency / analysability	+	+	–	+

Sampling-based planning is mature and well supported but is offline and, in its standard form, decoupled—disqualifying it as the primary whole-body planner while keeping it relevant as a baseline. Reactive QP-IK is fast and reactive but myopic. Learning-based control is attractive in principle but conflicts with the near-term, hardware-first, and analysability goals of this work. Whole-body MPC is the only family that simultaneously delivers coordinated base–arm motion, explicit constraint

handling, and online reactivity; its main drawback, the computational burden of solving an optimal-control problem online, is addressable through the MPC–MRT split and a distributed deployment (Chapter 3) and is quantified experimentally in Chapter 5.

Among whole-body MPC implementations, OCS2 is selected for its mature open-source implementation, its explicit constraint support, its demonstrated use on wheeled and legged mobile manipulators, and, decisive for a real-time deployment on distributed hardware, its process-separable MPC–MRT architecture. The remainder of this thesis therefore integrates and validates OCS2-based whole-body MPC on the HERo platform.

3

Integration of OCS2 into the HERo Mobile Manipulator

3.1 HERo Platform: Differential-Drive Base and UR10e Manipulator

3.2 Distributed System Architecture for Real-Time MPC

3.3 Whole-Body Model and Parameterization

3.4 Hardware Interfaces and ROS Control Layer Integration

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Simulation-Based Validation

4.1 Validation Methodology and Evaluation Metrics

4.2 Kinematic Closed-Loop Simulation (Dummy MRT)

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Hardware Deployment and Preliminary Experimental Results

5.1 Deployment on the HERo Platform

5.2 Deployment Challenges and Mitigations

5.3 Preliminary End-Effector Tracking Results

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Conclusion and Outlook

6.1 Summary

6.2 Critical Reflection

6.3 Recommendations for Resolving Present Issues

References

[BUCK22] A. F. Buckhorst, L. Grahn, and R. H. Schmitt. "Holistic Production Control for Line-less Mobile Assembly Systems". In: *Robotics and Computer-Integrated Manufacturing* (2022). VERIFY details.

[FARS17] F. Farshidian et al. *OCS2: An Open Source Library for Optimal Control of Switched Systems*. https://github.com/leggedrobotics/ocs2. Accessed: Friday 7th August, 2026. 2017.

[HÜTT16] G. Hüttemann, C. Gaffry, and R. H. Schmitt. "Adaptation of Reconfigurable Manufacturing Systems for Industrial Assembly – Review of Flexibility Paradigms, Concepts, and Outlook". In: *Procedia CIRP* (2016). VERIFY details.

[KERN19] W. Kern, F. Rusitschka, W. Kopp, and T. Bauernhansl. "Alternatives to Assembly Line Production in the Automotive Industry". In: *Proceedings of the International Conference on Production Research*. VERIFY details. 2019.

[MITT22] M. Mittal, N. Rudin, V. Klemm, A. Allshire, and M. Hutter. "Articulated Object Interaction in Unknown Scenes with Whole-Body Mobile Manipulation". In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. VERIFY details. 2022.

[PICK24] PickNik Robotics. *MoveIt 2: Motion Planning Framework for ROS 2*. https://moveit.ai. Accessed: Friday 7th August, 2026. 2024.

[RAWL17] J. B. Rawlings, D. Q. Mayne, and M. M. Diehl. *Model Predictive Control: Theory, Computation, and Design*. 2nd ed. Nob Hill Publishing, 2017.

[SAND22] T. Sandakalum and M. H. Ang Jr. "Motion Planning for Mobile Manipulators—A Systematic Review". In: *Applied Sciences* 12.21 (2022). VERIFY details.

[SANT26] T. Sant'Anna. *Systematic Literature Review – Methodology Report: Whole-Body Motion Planning and Control Integration in Robotic Navigation Stacks*. Systematic Literature Review Report. Unpublished internal report, KALIMORO project. Laboratory for Machine Tools and Production Engineering (WZL), RWTH Aachen University, 2026-04.

[SLEI21] J.-P. Sleiman, F. Farshidian, M. V. Minniti, and M. Hutter. "A Unified MPC Framework for Whole-Body Dynamic Locomotion and Manipulation". In: *IEEE Robotics and Automation Letters* 6.3 (2021). VERIFY details.

[THAK23] S. Thakar, S. Srinivasan, S. Al-Hussaini, et al. “A Survey of Wheeled Mobile Manipulation: A Decision-Making Perspective”. In: *Journal of Mechanisms and Robotics* (2023). VERIFY details.

[UNIV24] Universal Robots. *UR10e Collaborative Industrial Robot Arm*. https://www.universal-robots.com. Accessed: Friday 7th August, 2026; VERIFY payload/reach specs. 2024.